

Damage Review Copilot

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Vision

Claims agents spend most of their day looking at photos of dented cars and typing up estimates. It is slow, repetitive, and expensive, and quality depends on whoever picks up the claim. "Damage Review Copilot" is an internal prototype that changes what the agent's job is: instead of building every estimate from scratch, the AI drafts the assessment and the agent reviews it, item by item, with the evidence and cost sources in front of them.

The goal is to cut handling time and cost per claim without weakening the thing that makes the process trustworthy: a human accountable for every number. The AI proposes -> the agent decides -> a senior adjuster authorises.

Coverage, fraud, denial, and payment decisions are out of scope, as is anything a photo cannot prove; hidden or structural damage always routes to a person.

The riskiest assumption V1 only helps if enough real claims arrive with usable photos and simple visible damage. We do not know that percentage yet. So the pilot starts in shadow mode: the AI drafts assessments on live claims while agents work exactly as they do today, and we measure what share of claims it could actually have helped before anyone relies on it.

User Stories

1. As a claims agent, I want a queue ordered by submission time and age, with assessments running in the background, so I work the oldest claim instead of waiting on the AI.
2. As a claims agent, I want to judge the photos myself before seeing any AI output, so I never build an estimate on evidence too blurry or dark to trust.
3. As a claims agent, I want to confirm, correct, or reject each AI finding one at a time, so nothing reaches the estimate without my explicit sign-off.
4. As a claims agent, I want every repair operation and price to cite its source, so I can verify a number instead of taking it on faith.
5. As a claims agent, I want to add findings the AI missed, request better photos, or route the whole claim to manual review in one step.
6. As a senior adjuster, I want a handoff packet showing what the AI proposed, what the agent changed and why, and what the photos could not show, so I can authorise repairs with full context.

Key Features

All six are built and demonstrable in the prototype.

Feature	What it does
Claims evaluation queue	Asynchronous dashboard ordered by submission time and age, showing evidence, eligibility, estimate state, and next action. Assessments run in the background; the agent applies updates when ready.
Evidence gate	Before any AI annotation appears, the agent inspects the raw photos and makes a decision: adequate, request more photos, or route to manual review. Annotation is off by default.
Sequential finding review	Findings are presented one at a time. Each requires Confirm, Correct (with a reason and source), or Reject (with a reason). There is no bulk accept.
Cited estimates	Every operation and price links to a source excerpt from the repair guide and pricebook. Retrieval-backed in production; clearly labelled fixtures in the prototype.
Two-track estimate	The AI's proposal is immutable and shown beside the agent-confirmed subtotal. Unresolved findings never count towards the reviewed total.
Senior handoff packet	Decision counts, estimate delta, retained limitations, and the full audit trail travel to the senior adjuster, who alone authorises repairs.

Prioritisation

I prioritised against the riskiest failure mode first: automation bias. An early version showed the whole AI draft on one page with a single acknowledgement checkbox. An agent under time pressure would scroll past everything and tick the box, which makes the tool worse than the manual process, just faster at being wrong. I rebuilt the flow so each finding needs its own decision and the evidence gate forces a human judgement before AI output is even visible. Those controls are P0.

The queue is P0 because model latency is real and agent idle time is the most direct cost. Citations are P0 because an unverifiable number gets either blind-trusted or ignored, and both defeat the product.

Deferred: policyholder-facing intake, video evidence, fraud signals, auto-settlement of low-value claims, and claims-system integrations.

Success Metrics

Efficiency: average handling time per claim against the pre-launch baseline; claims per agent per day.

Accuracy: gap between the approved estimate and the final negotiated repair cost; senior return, supplement, and reopen rates.

Human-AI interaction: the confirm/correct/reject mix and evidence-gate rejection rate: a confirm rate near 100% is a warning sign of rubber-stamping, so it gets monitored.

Model quality: precision and recall on the expert-adjudicated benchmark, with error rates sliced by vehicle type, lighting, and image quality.

Guardrail: missed safety-critical damage must not rise above the manual baseline; this is the go/no-go gate (see AI Integration).

Pilot hypothesis: 30% lower handling time with 100% item-level human review and no guardrail regression.

What the bet is worth (illustrative). Placeholder numbers, to be replaced during baseline discovery: a mid-size insurer handling 200,000 eligible claims a year, where an agent spends about 45 minutes actively reviewing each one at a fully loaded cost (salary plus overheads) of about \$40 an hour, spends roughly \$30 per claim — around \$6m a year — on this step alone. The 30% saving we are betting on is worth about \$9 per claim, or \$1.8m a year. Running the AI — model inference, retrieval, audit storage, review tooling — should cost on the order of \$1 per claim, about \$200k a year, leaving roughly \$1.6m net.

AI Integration

Production pipeline. We would not train models from scratch: the plan is to fine-tune a strong pre-trained vision model on the insurer's own adjudicated claim photos, because the hard part is damage-specific judgement, not general image understanding. An evidence-quality model checks blur, lighting, and coverage first; insufficient evidence pauses the claim with no findings and no estimate. For adequate evidence, the vision model outputs visible-damage observations in a fixed, canonical vocabulary: part, damage type, severity, image region, rather than free text, so every downstream step keys off the same terms. Repair guidance may use retrieval over the insurer's approved repair guides. Pricing is never generated by a model: it is a deterministic lookup keyed by VIN/trim, part, operation, geography, labour class, and pricebook version, so the same damage always prices the same, and every number is traceable. The thresholds that decide when evidence is good enough, and when a finding is confident enough to show, are calibrated against expert-adjudicated data. And the system abstains, no findings, straight to a human, when images conflict, when a case looks unlike anything in its training data, or when there is any sign of structural involvement.

Evaluation and the go/no-go gate. Before launch, we build a benchmark of real claim photos adjudicated by senior estimators, and every model version is scored against it. Most metrics are targets; one is a gate: missed safety-critical damage. If the model misses dangerous damage the experts caught, it does not ship, whatever else improved. Agent corrections enter training or evaluation data only after expert adjudication, never straight from interaction logs.

Human-AI interaction. The contract is simple: the AI may propose, the human decides. The AI flags evidence gaps, suggests findings, retrieves costs, and states its limitations. The agent decides whether evidence is sufficient, disposes of every finding individually, can add findings the AI missed, and decides when the packet is ready. The senior adjuster authorises or returns it. Original AI suggestions, agent reasons, and before/after values are preserved in the audit trace, so accountability survives the review.

Ethical considerations. Automation bias is handled by design (evidence-first review, per-item decisions, no bulk accept) and watched through the confirm-rate metric. Hidden-damage risk is handled with narrow eligibility, the persistent limitations panel, and human routes that are always one click away. Fairness is monitored by slicing error rates across vehicle types, lighting, and device quality, since a model that only works on new cars in daylight quietly disadvantages some policyholders. Suspicious or manipulated media routes to human review rather than automatic denial, and unclear cases always reach a person with the power to help.